Quantum Computing

Chapter 00: Introduction

Prachya Boonkwan

Version: August 3, 2026

Sirindhorn International Institute of Technology
Thammasat University, Thailand

Who? Me?

- Nickname: Arm (P'/N' Arm, etc.)
- Born: Aug 1981
- Work
  - Researcher at NECTEC 2005-2024
  - Lecturer at SIIT, Thammasat University 2025-now
- Education
  - B.Eng & M.Eng in Computer Engineering, Kasetsart University, Thailand
  - Obtained Ministry of Science and Technology Scholarship of Thailand in early 2008
  - Did a PhD in Informatics (AI & Computational Linguistics) at University of Edinburgh, UK from 2008 to 2013

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Course Outline

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- Lec 6: State measurement
- Lec 7: Deutsch-Jozsa algorithm and phase kickback technique
- Midterm exam

Second Half

- Lec 8: Quantum noisy channels
- Lec 9: Grover's search, Fourier transform, and phase estimation
- Lec 10: Shor's algorithm
- Lec 11: Quantum simulation
- Lec 12: Variational quantum algorithms
- Lec 13: Quantum machine learning
- Lec 14: Quantum optimization
- Final exam

Grading Scheme

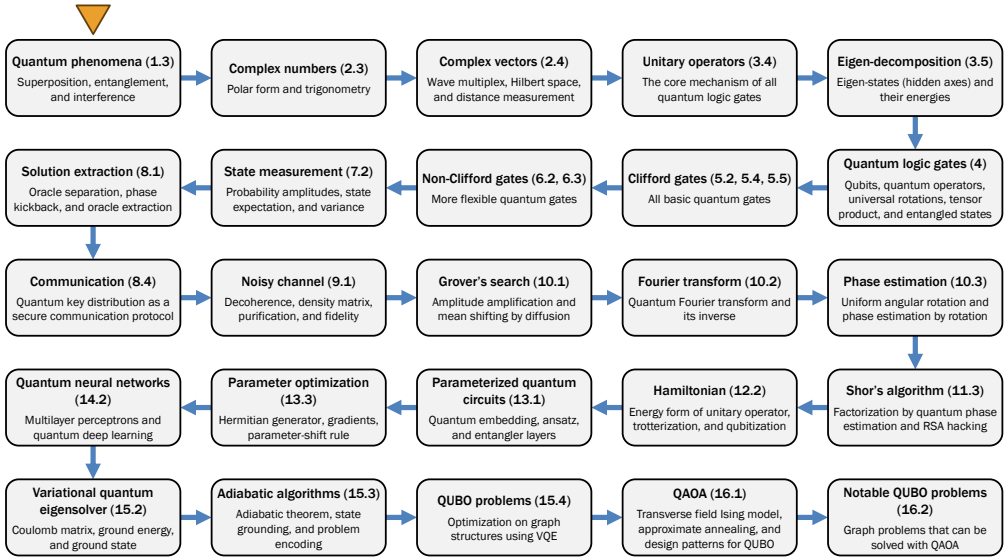
- Midterm and final exams
  - Open-slides, open-book, and open-calculator
  - Majority of my questions are based on discussion and reasoning, while only a few are calculation-oriented (i.e. virtually giving free scores)
- Assignments
  - Two take-home assignments based on calculation and quantum programming on simulators
- No-F's policy: minimum grade is D+
- Github repository for the course is `https://github.com/kaamanita/quantum-computing`
- Odorless foods and drinks are allowed in the class due to proximity to lunch time

Items	Proportions
Midterm	35%
Final	35%
Assignments	20%
Attendance	10%
TOTAL	100%

My Forthcoming Book

"Quantum Computing: A Gentle Introduction for AI Engineers" (Boonkwan, 2026)

Quantum Topics You Really Need to Know



Classical vs. Quantum Computers

What the (quantum) fuss is all about?

- Hearsay: “Quantum computer is immensely faster than classical computer!”
  - Shor’s algorithm for factoring an integer n in $O((\log n)^3)$ time complexity, making RSA code decryption feasible in polynomial time
  - Quantum supremacy vs. quantum advantage
- Quantum computers
  - Perform computation by manipulating state superposition and entanglement in-memory with quantum-mechanical phenomena
  - Are susceptible to noise due to quantum interference
  - Can solve some polynomial-time and non-polynomial-time problems with bounded error
- The notion of classical computer refers to modern-day computers that rely on binary data

Figure 1: Quantum computer

Classical Computer

- Data: Stored as binary numbers and manipulated by electronic circuits
- Processing: Logical and arithmetic operations, data transfer, and type conversion
- Representation: Only one item out of a vast realm of all possible configurations!
- Limit: We cannot intrinsically process multiple input items in parallel

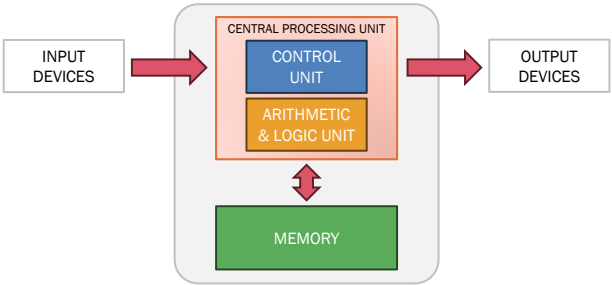


Figure 2: von Neumann architecture of classical computers

Bitstring

- Binary number is a number expressed in the base-2 numeral system: 0 and 1
- Bitstring (or binary representation) of integer N is denoted by $\mathbb{B}_L(N)$, where L is the length of such bitstring, e.g.

$$\mathbb{B}_8(33) = 00100001_2$$

- We can convert a bitstring to a decimal number by multiplying each k-th digit (from the right) with its decimal equivalent 2^{k-1} and summing these multiplicants, e.g.

$$\begin{aligned} 00100001_2 &= 1 \times 2^5 + 1 \times 2^0 \\ &= 33_{10} \end{aligned}$$

- We can also convert an integer to a bitstring by continued division as follows
  1. Find the closest exponent 2^k to N
  2. Subtract 2^k from N
  3. Repeat until the subtraction becomes zero
- Example: We convert 30_{10} to 11110_2 by

$$\begin{aligned} 30 &= 1 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 \\ &\quad + 1 \times 2^1 + 0 \times 2^0 \end{aligned}$$

2	30	14	6	2	0
	-16	-8	-4	-2	
	2^4	2^3	2^2	2^1	

Table 1: Conversion from 30_{10} to 11110_2

Logical Operations

- AND returns 1 iff both operands are 1

$$0011 \wedge 0101 \quad = \quad 0001$$

- OR returns 1 iff at least one operand is 1

$$0011 \vee 0101 \quad = \quad 0111$$

- NOT returns the opposite value

$$\overline{01} \quad = \quad 10$$

- XOR returns 1 iff only one operand is 1

$$0011 \oplus 0101 \quad = \quad 0110$$

- These abstract logical operations lay the foundation of data processing in classical computing, where conditional branching, jumping, and looping are predominant
- In quantum computing, these operations will be emulated by linear transformation

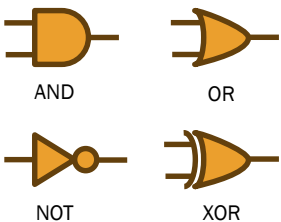


Figure 3: Logical operations

Limitations of Classical Computing

- Classical computing is linearly scalable, due to sequential logical processing of discrete states
- Parallelism in classical computing accelerates sequential processing with multiple computing cores

```
import multiprocessing as mp

def factorial(x):
    result = 1
    for i in range(2, x+1):
    result *= i
    return result

items = [10, 20, 30, 40, 50, 60, 70, 80, 90, 100]
pool = mp.Pool(processes=4)
results = pool.map(factorial, items)
print(results)
```

- Classical parallelism struggles on looping with exponential iterations, resulting in $O(2^N)$ time complexity
- In quantum computing, multiple discrete values are represented as a superposition of bitstrings

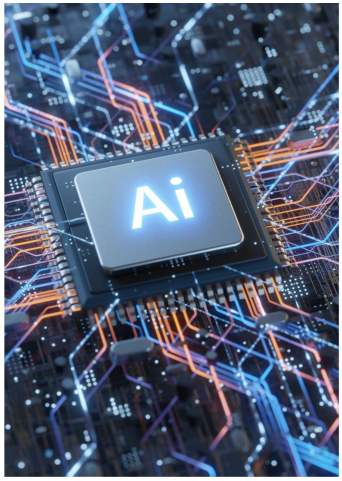
$$\{0000, 0001, \ldots, 1111\} \quad \Rightarrow \quad \{0,1\}^4$$

Then the quantum operations are done on this state superposition

- The quantum data processing is based on complex algebra (i.e. complex vectors, complex matrices, and measurement)

Modern-Day AI

- High demand of parallel computation with graphical processing units (GPUs)
- Limitations of high perforamance computing
  - Power wall: Performance gain of transistor-based CPUs are limited by thermal ceiling
  - Memory wall: High bandwidth memory still lags behind when compared to the CPUs
  - Network bottleneck: Data synchronization between computing units are relatively very slow
  - Quantum tunneling: Electrons start to leak through insulation across transistors causing errors & heat



- Nondeterminism: probabilistic information processing via quantum mechanics (but we don't have to deal with it)

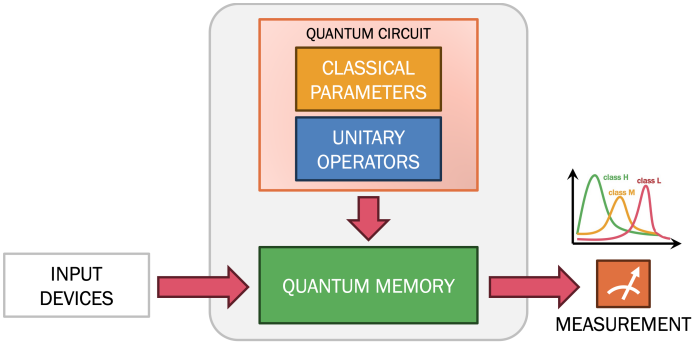


Figure 4: Quantum architecture

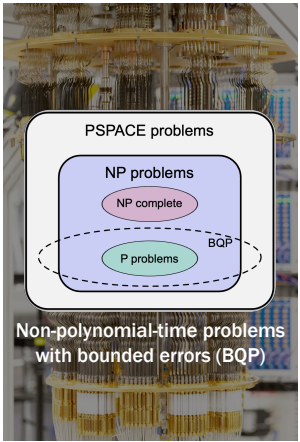


Figure 5: NP-time problems with bounded errors (BQP)

Quantum Mechanics

Quantum Memory

- Qubit: Each bit of information is stored as the momentum spin of a particle
- Information processing is to manipulate these qubits via physical interaction e.g. laser beam (trapped ions) or light interference (photons)

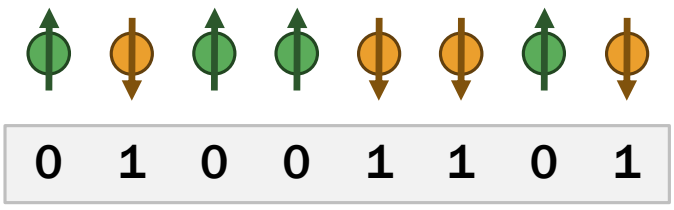


Figure 6: A classical bitstring represents the system's current state.

State Measurement

- Momentum spins are destroyed when we read the system's current state

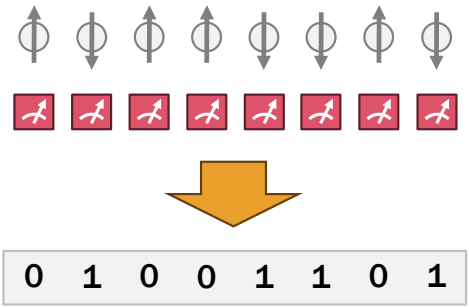
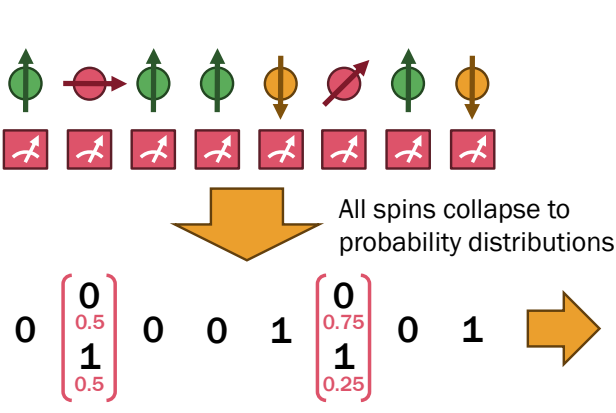


Figure 7: State measurement

Phenomenon 1: Superposition

- A quantum system can exist in multiple states (with probabilities) in parallel



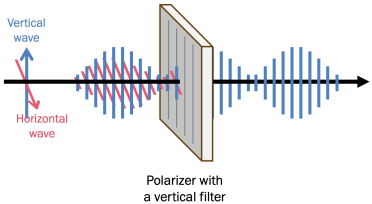


Figure 8: Quantum superposition

Quantum State and Probability Amplitudes

- Quantum state: the system's configuration of momentum spins for each qubit
- Three dipoles (0/1, $+/-$, and $+i/-i$) represent the 3D momentum spin of one qubit

$$|\psi\rangle \quad = \quad a\,|0\rangle + b\,|1\rangle$$

where complex numbers a and b are probability amplitudes of $|\psi\rangle$ collapsing to 0 and 1

- Probability density of $|\psi\rangle$ collapsing to 0 or 1:

$$P(0|\psi) = a^2 \qquad P(1|\psi) = b^2$$

- Dirac's notation: $|\psi\rangle$ reads 'ket'-psi

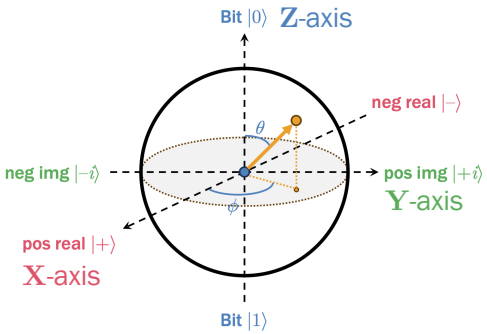


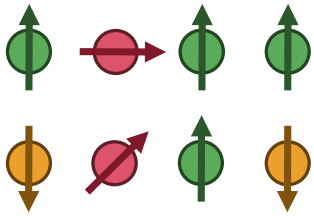
Figure 9: Bloch sphere

Multi-Qubit System

- Tensor product of quantum states

$$\left(a_1|0\rangle + b_1|1\rangle\right) \otimes \left(a_2|0\rangle + b_2|1\rangle\right)$$
$$= a_1a_2|00\rangle + a_1b_2|01\rangle + b_1a_2|10\rangle + b_1b_2|11\rangle$$

where $\otimes$ denotes concatenation



- Factorization of quantum state

$$\frac{\sqrt{3}}{2\sqrt{2}}|00001001\rangle + \frac{1}{2\sqrt{2}}|00001101\rangle + \frac{\sqrt{3}}{2\sqrt{2}}|01001001\rangle + \frac{1}{2\sqrt{2}}|01001101\rangle$$

Multi-state superposition

$$|0\rangle \otimes \left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right) \otimes |0\rangle \otimes |0\rangle \otimes |1\rangle \otimes \left(\frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle\right) \otimes |0\rangle \otimes |1\rangle$$

Tensor product

Dirac's Notation

- Single-qubit ket = 2-dim column vector

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad a|0\rangle + b|1\rangle = \begin{bmatrix} a \\ b \end{bmatrix}$$

- L-qubit ket = 2^L-dim column vector

$$|k\rangle = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \qquad \sum_{k=0}^{2^L-1} a_k |k\rangle = \begin{bmatrix} a_0 \\ \vdots \\ a_{2^L-1} \end{bmatrix}$$

- $|k\rangle$ is 1-hot vector at the k-th position

- Single-qubit bra = 2-dim row vector

$$\Big(a|0\rangle + b|1\rangle\Big)^* = \overline{a}\langle 0| + \overline{b}\langle 1| = \begin{bmatrix} \overline{a} & \overline{b} \end{bmatrix}$$

- L-qubit bra = 2^L-dim row vector

$$\sum_{k=0}^{2^L-1} a_k \langle k| = \begin{bmatrix} a_0 & \dots & a_{2^L-1} \end{bmatrix}$$

- Bra-ket product = vector similarity

$$\langle k|k\rangle = 1 \quad \langle j|k\rangle = 0$$

where $j \neq k$

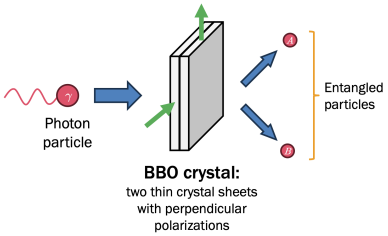
Phenomenon 2: Entanglement

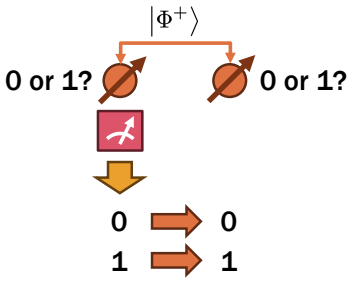
- Multiple particles may become inextricably linked to each other, i.e. they are entangled
- Entangled states are multi-qubit states that cannot be factorized further, e.g. Bell states

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} \qquad |\Phi^-\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}}$$

- If one of them collapses, the rest of the pack also collapse instantaneously:

$$P(00|\Phi^+) = \frac{1}{2} \qquad P(11|\Phi^+) = \frac{1}{2}$$





Phenomenon 3: Quantum Interference

- If we manipulate the prob amplitude of a qubit, this change ripples to the others
- Probability amplitude can be manipulated in two ways
  - Amplification: increasing the prob amplitude
  - Dampening: decreasing the prob amplitude
- Quantum interference is the key mechanism of quantum gate design

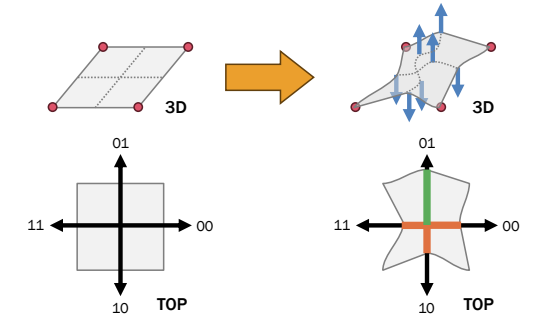


Figure 10: Quantum interference

Quantum Operations

Designing a Quantum Operator/1

- We manipulate the system's quantum state by applying a quantum operator
- State mapping

$$\begin{aligned} |\text{input } \psi_1\rangle &\mapsto |\text{output } \phi_1\rangle \\ &\vdots \\ |\text{input } \psi_K\rangle &\mapsto |\text{output } \phi_K\rangle \end{aligned}$$

- Quantum operator

$$\mathbf{U} = \sum_{k=1}^{K} |\text{output } \phi_k\rangle \langle \text{input } \psi_k|$$

- Reminder: $\langle\psi| = |\psi\rangle^*$

- Generally, we use pure states as inputs ψ_k
- The quantum operator applies interference on the target qubits

$$\begin{aligned} &\mathbf{U}\,|\text{state } \xi\rangle \\ &= \left(\sum_{k=1}^{K} |\text{output } \phi_k\rangle \langle \text{input } \psi_k| \right) |\text{state } \xi\rangle \\ &= \sum_{k=1}^{K} |\text{output } \phi_k\rangle \underbrace{\left(\langle \text{input } \psi_k | \text{state } \xi\rangle \right)}_{\text{similarity}} \\ &= \sum_{k=1}^{K} |\text{output } \phi_k\rangle \langle \text{input } \psi_k | \text{state } \xi\rangle \end{aligned}$$

Designing a Quantum Operator/2

- Example: NOT operator has state mapping

$$|0\rangle \mapsto |1\rangle \qquad |1\rangle \mapsto |0\rangle$$

The quantum operator for NOT is

$$\begin{aligned} \mathrm{X} &= |1\rangle\langle 0| + |0\rangle\langle 1| \\ &= \begin{bmatrix} 0 \\ 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix}\begin{bmatrix} 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \end{aligned}$$

- Compared with matrix multiplication

$$\begin{aligned} \mathrm{X}\begin{bmatrix} a \\ b \end{bmatrix} &= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} a \\ b \end{bmatrix} \\ &= \begin{bmatrix} 0a + 1b \\ 1a + 0b \end{bmatrix} \\ &= \begin{bmatrix} b \\ a \end{bmatrix} \end{aligned}$$

- Compared with state mapping

$$\begin{aligned} \mathrm{X}\Big(a|0\rangle + b|1\rangle\Big) &= a\Big(\mathrm{X}|0\rangle\Big) + b\Big(\mathrm{X}|1\rangle\Big) \\ &= a|1\rangle + b|0\rangle \\ &= b|0\rangle + a|1\rangle \end{aligned}$$

NOT Gate: X

- X gate flips the qubit between $|0\rangle$ and $|1\rangle$
- State mapping

$$|0\rangle \mapsto |1\rangle \qquad |1\rangle \mapsto |0\rangle$$

- It imitates logical NOT in classical computing

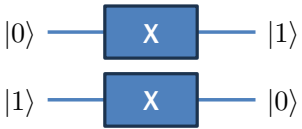
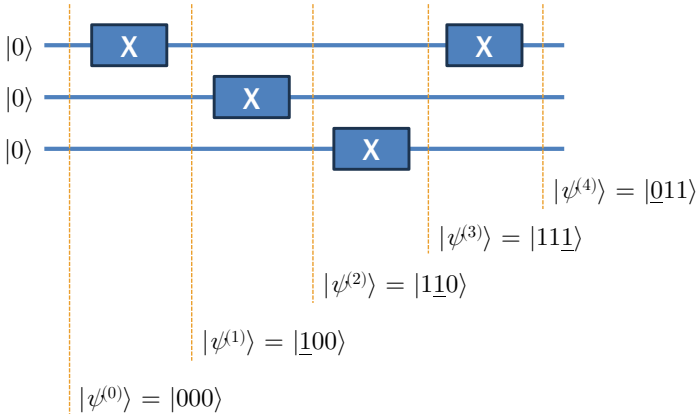


Figure 11: X gate

- Quantum algorithm is a circuit of quantum logic gates



Hadamard Gate: H

- H gate flips between pure states and superpositions
- State mapping

$$|0\rangle \mapsto \underbrace{\frac{|0\rangle + |1\rangle}{\sqrt{2}}}_{|+\rangle} \qquad |1\rangle \mapsto \underbrace{\frac{|0\rangle - |1\rangle}{\sqrt{2}}}_{|-\rangle}$$

- Used for parallel processing

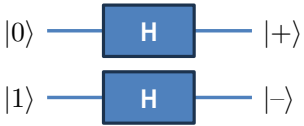
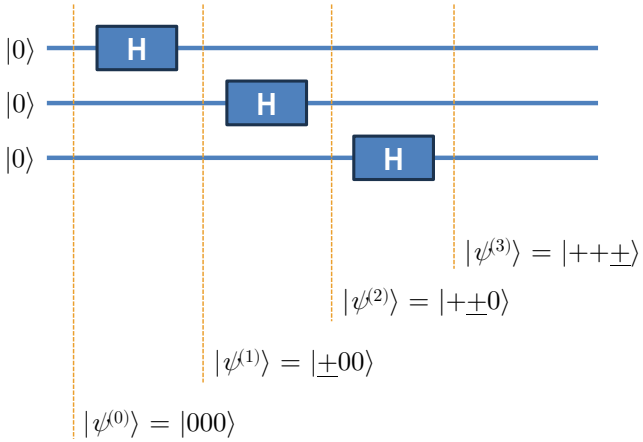


Figure 12: Hadamard gate

- Preparing a superposition of all possible 2^L inputs



Controlled NOT Gate: CNOT

- CNOT gate imitates an if-clause and creates entanglement
- State mapping

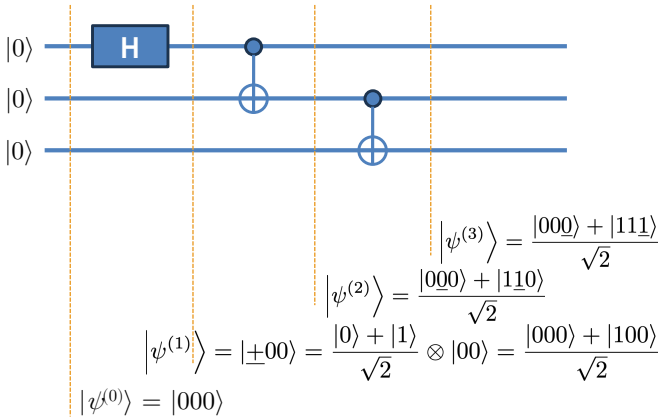
$$|0x\rangle \mapsto |0x\rangle \qquad |1x\rangle \mapsto |1\bar{x}\rangle$$

where x is either 0 or 1

- Last qubit is the output equivalent to classical XOR gate

Figure 13: Controlled NOT gate

- Preparing an entangled state with H and CNOT gates



Phase Shift Gate: S

- S gate flips between real and complex numbers on $|1\rangle$
- State mapping

$$|0\rangle \mapsto |0\rangle \qquad |1\rangle \mapsto i\,|1\rangle$$

- Imaginary number $i = \sqrt{-1}$ is used as soft feature in quantum machine learning

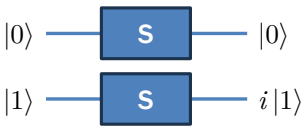
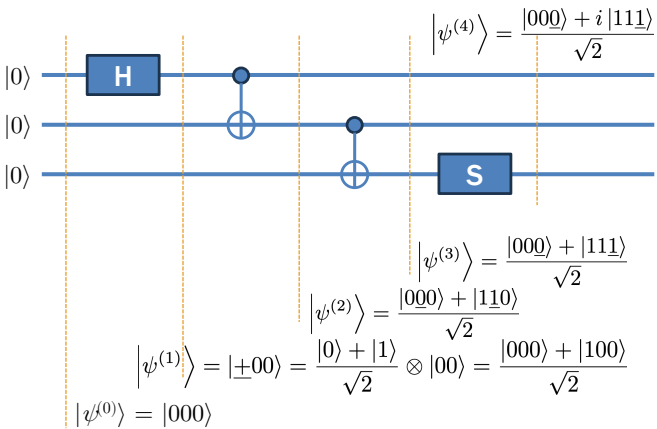


Figure 14: Phase shift gate

- Preparing an entangled state with complex probability amplitudes



Toffoli Gate: CCNOT

- CCNOT gate imitates an if-clause with two control qubits
- State mapping

$$\text{Both controls: } |11x\rangle \mapsto |11\bar{x}\rangle$$
$$\text{Otherwise: } |xyz\rangle \mapsto |xyz\rangle$$

where x and y are binary flags

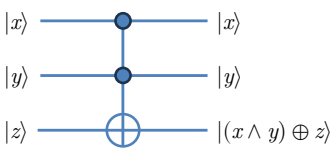
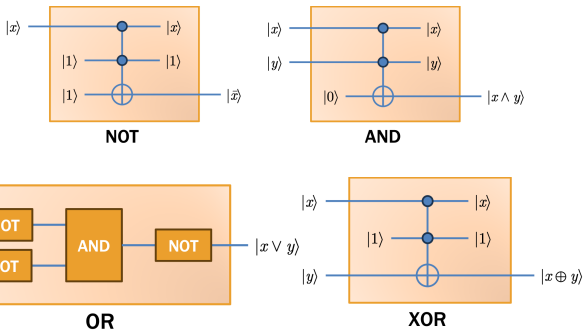


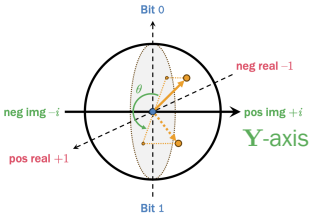
Figure 15: Toffoli gate

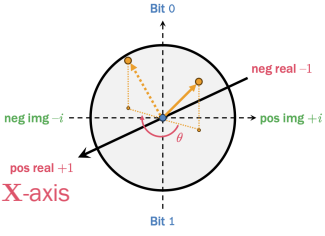
- CCNOT gate imitates all classical logic gates, allowing massive parallelism on classical programs

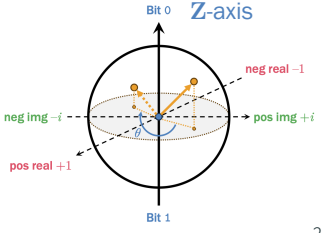


Rotation Gates: RX, RY, and RZ

- Rotation gates rotates a qubit around the specified eigenbases
- Rotation alters the probability densities used in quantum machine learning
- X-rotation $\mathbf{RX}(\theta)$, Y-rotation $\mathbf{RY}(\theta)$, and Z-rotation $\mathbf{RZ}(\theta)$, where θ is a model parameter







Controlled Gate: $\mathbf{C}^{|k\rangle}\mathbf{U}$

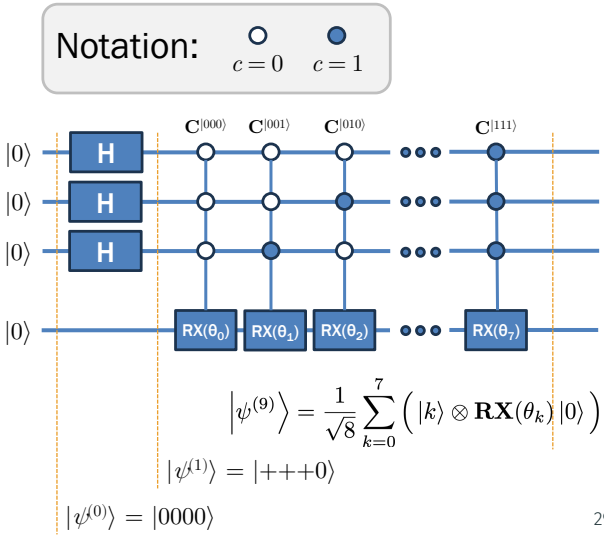
- Extension of CNOT, where control qubits are $|k\rangle$
- State mapping of $\mathbf{C}^{|k\rangle}\mathbf{U}$
  - If controls $|x_1, \ldots, x_L\rangle = |k\rangle$

$$|x_1, \ldots, x_L, y\rangle \mapsto |x_1, \ldots, x_L\rangle \otimes \mathbf{U}\,|y\rangle$$

  - Otherwise

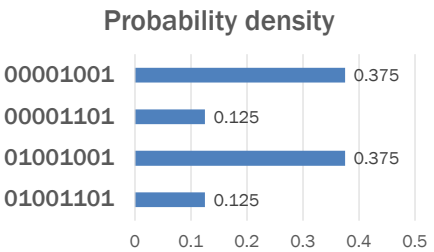
$$|x_1, \ldots, x_L, y\rangle \mapsto |x_1, \ldots, x_L, y\rangle$$

- This is frequently used in Fourier transform and phase estimation



State Measurement

- Probability density: We destructively measure the probability distribution of each pure state by projection (i.e. the quantum state collapses)



- Expectation: We non-destructively measure if each k-th qubit leans towards one side of the particular dipoles $\mathbf{U} \in \{\mathbf{X}, \mathbf{Y}, \mathbf{Z}\}$

$$\left\langle \mathbf{U}^{(\psi)} \right\rangle = \langle \psi | \ \mathbf{U} \ | \psi \rangle$$

Expectations	Positive (+1)	Negative (−1)
$\langle \mathbf{Z}^{(k)} \rangle$ or Z_k	Bit 0	Bit 1
$\langle \mathbf{X}^{(k)} \rangle$ or X_k	Pos real	Neg real
$\langle \mathbf{Y}^{(k)} \rangle$ or Y_k	Pos img	Neg img

Quantum Algorithms

Oracle Extraction

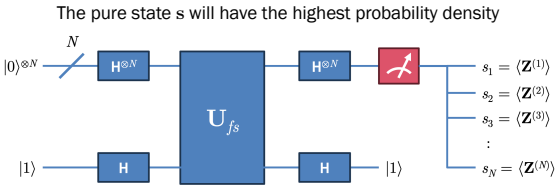
- Bernstein-Vazirani algorithm
  - Extracting an oracle **s** from a black-box function f_s by converting it to $\mathbf{U}_{f_s}$
- Phase kickback technique
  - State mapping of $\mathbf{U}_{f_s}$

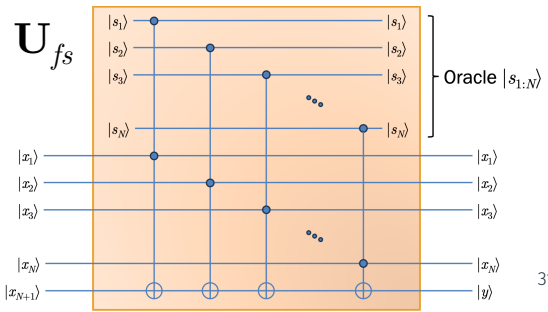
$$\text{Oracle: } |s_{1:N}, y\rangle \mapsto |s_{1:N}, y \operatorname{cis}\theta\rangle$$
$$\text{Otw.: } |x_{1:N}, y\rangle \mapsto |x_{1:N}, y\rangle$$

  - Kickback property

$$\mathbf{U}_{f_s}\left(|+\rangle^{\otimes N} \otimes |-\rangle\right) = \left(\bigotimes_{k=1}^{N} \frac{|0\rangle + \left(\operatorname{cis}\theta\right)^{f_{sk}(2^{k-1})}|1\rangle}{\sqrt{2}}\right) \otimes |-\rangle$$

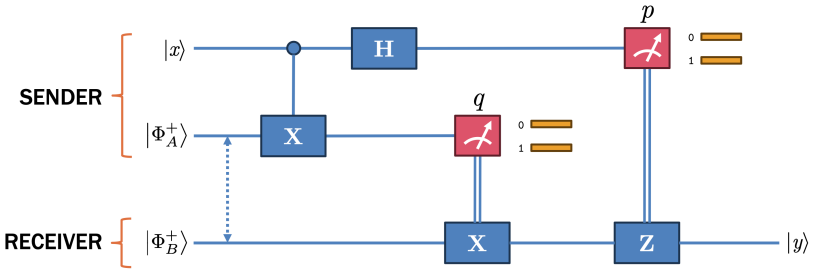
  - The last qubit interferes the others





Quantum Teleportation

- Secure communication based on quantum mechanics
- We want to transmit state $|x\rangle = a\,|0\rangle + b\,|1\rangle$ with entangled state $\Phi^+ = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$
- How it works:
  1. The sender and receiver have $\left|\Phi_A^+\right\rangle$ and $\left|\Phi_B^+\right\rangle$, respectively
  2. The sender runs the quantum teleportation algorithm on the input qubit $|x\rangle$
  3. The sender measures p and q and send them over classical communication
  4. The receiver uses them to translate $\left|\Phi_B^+\right\rangle$ to $|x\rangle$



p	q	State $\|y\rangle$
0	0	$a\,\|0\rangle + b\,\|1\rangle$
0	1	$b\,\|0\rangle + a\,\|1\rangle$
1	0	$a\,\|0\rangle - b\,\|1\rangle$
1	1	$-b\,\|0\rangle + a\,\|1\rangle$

Quantum Key Distribution

- Secure communication without actual key transmission, such as BB84

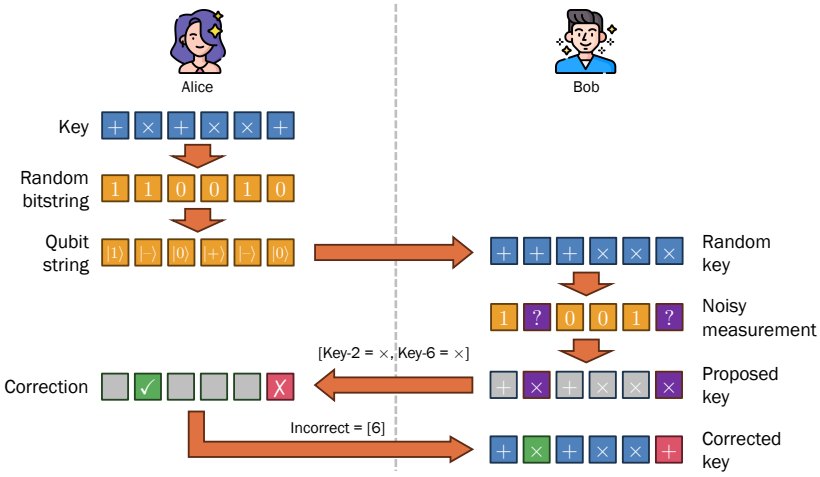


Figure 16: BB84 Protocol

Grover's Search Algorithm

- We can search for 1 of N items w.r.t. the oracle function f with $k^* = \left\lfloor \frac{\pi}{4}\sqrt{N} \right\rfloor$ iterations

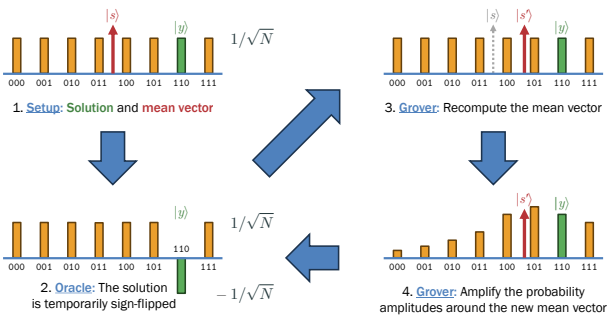


Figure 17: Overview of Grover's algorithm

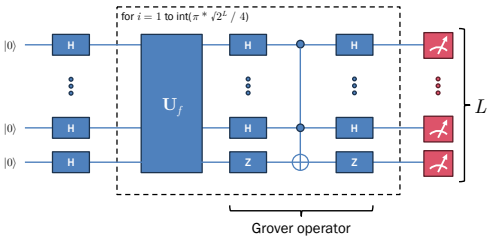
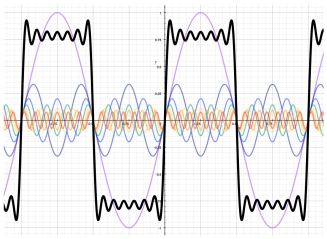
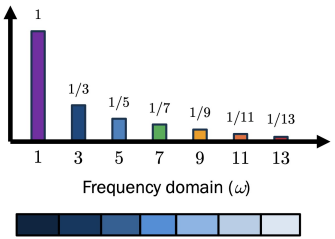


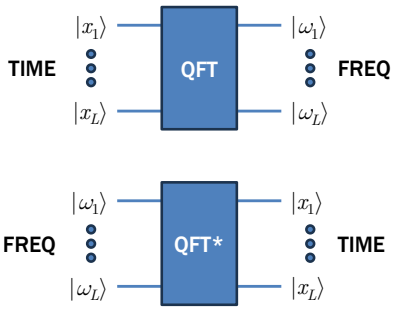
Figure 18: Grover's algorithm. $\mathbf{U}_f$ is equivalent to Step 2, while the Grover's operator is equivalent to Steps 3 and 4.

Quantum Fourier Transform

- Translates a time series of N items to a frequency domain, and vice versa, with $O((\log N)^2)$ time complexity



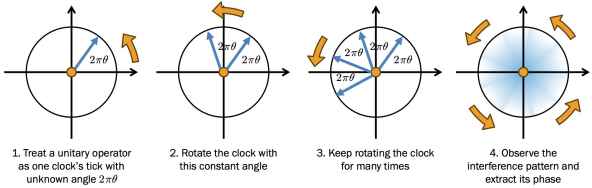


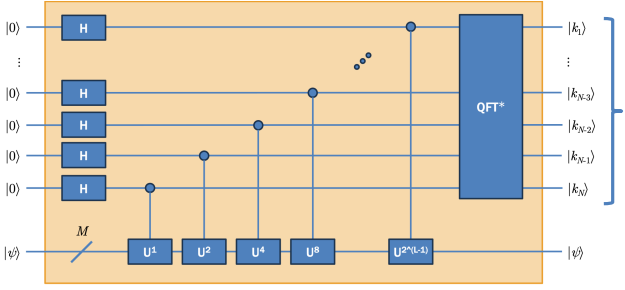


where $L = \log N$

Quantum Phase Estimation

We can extract a phase $2\pi\theta$ of a black-box algorithm $\mathbf{U}$ by rotating it in different frequencies and observe the resonance frequency





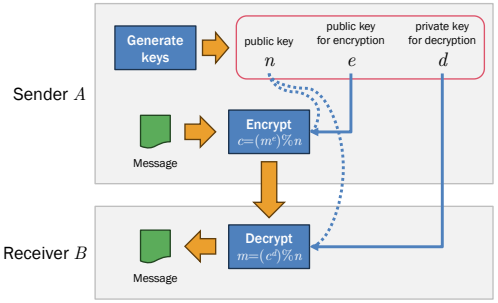
Measure for k^* = pure state with max prob

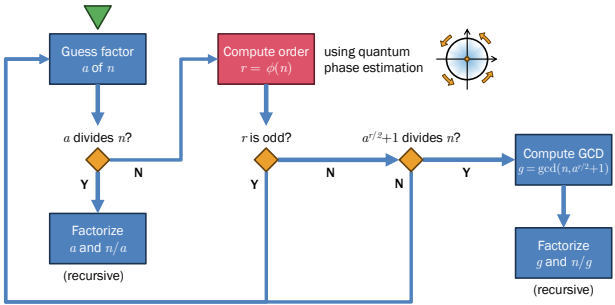
Phase $\theta = k^*/2^N$

Quantum-Suitable Problems

Shor's Algorithm

- RSA cryptosystem
  - Public and two private keys (n, e, d)
  - Encryption: $c = (m^e)\%n$
  - Decryption: $m = (c^d)\%n$
- Attack with QPE
  - Modulo Umod n(a) is eqv. to rotation
  - QPE is used to guess the phase
  - Time complexity $O(L^2 \log L)$





Simulation with Trotterization

- Hamiltonian $\hat{\mathbf{H}}_{\mathbf{U}}$ is the energy form of quantum circuit $\mathbf{U}$ of L qubits

$$\mathbf{U} = \mathrm{cis}\left(-\hat{\mathbf{H}}_{\mathbf{U}}\right)$$

- We can compute $\hat{\mathbf{H}}_{\mathbf{U}}$ by

$$\hat{\mathbf{H}}_{\mathbf{U}} = \sum_{\mathbf{P}\in\{\mathbf{I},\mathbf{X},\mathbf{Y},\mathbf{Z}\}^{\otimes L}} \left[\mathbf{P}\times \mathrm{trace}(\mathbf{UP})\right]$$

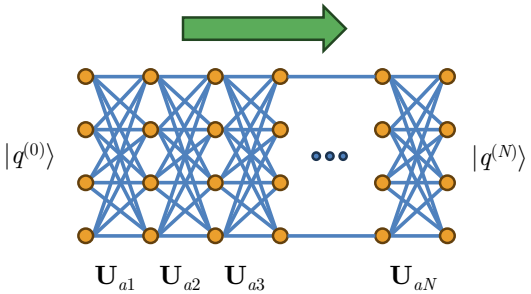
where $\mathbf{I},\mathbf{X},\mathbf{Y},\mathbf{Z}$ are Pauli matrices

$$\mathbf{I}=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix} \quad \mathbf{X}=\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}$$

$$\mathbf{Y}=\begin{bmatrix}0 & -i\\ i & 0\end{bmatrix} \quad \mathbf{Z}=\begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}$$

- Trotter-Suzuki decomposition: We can simulate N iterations of $\mathbf{U}$ by approximation

$$\mathbf{U}^N \approx \lim_{n\to\infty}\left[\mathrm{cis}\left(-\frac{N}{n}\hat{\mathbf{H}}_{\mathbf{U}}\right)\right]^n$$



- Variational quantum algorithm
  - Quantum circuit with rotation gates $\mathbf{RX}(\theta)$, $\mathbf{RY}(\theta)$, and $\mathbf{RZ}(\theta)$
  - Basic entangler layer ($\mathbf{BEL}$)

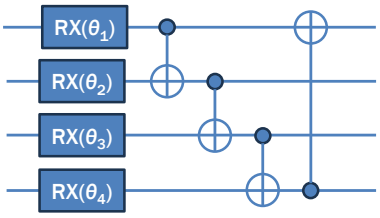


Figure 19: Basic entangler layer

- Quantum embedding $\Lambda(\Theta)$
  - Angle embedding: params for $\mathbf{RX}(\theta_k)$
  - Amplitude embedding: params for each pure state
- Hybrid classical-quantum computing

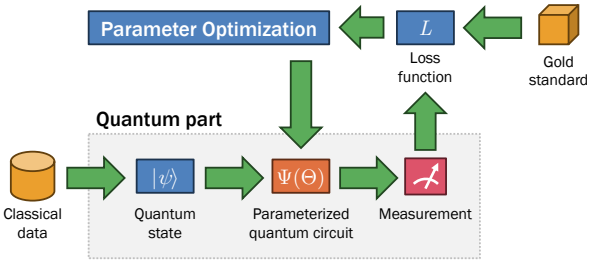
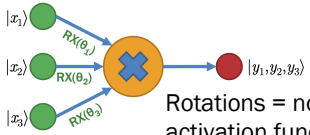


Figure 20: Hybrid system

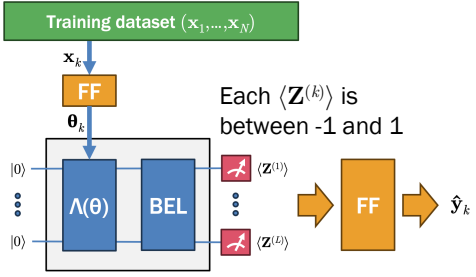
Quantum Neural Networks

- Quantum perceptron



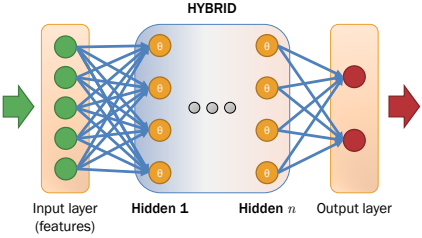
Rotations = nonlinear activation functions

- Quantum multilayer perceptrons

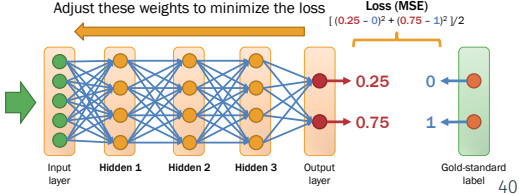


Each $\langle \mathbf{Z}^{(k)} \rangle$ is between -1 and 1

- Quantum deep NNs



- Backpropagation



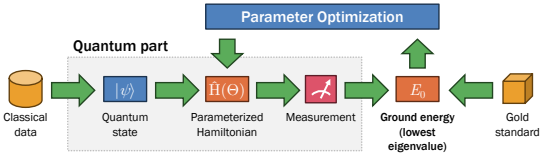
Variational Quantum Eigensolver (VQE)

- We want to minimize a loss Hamiltonian defined on qubit expectations

$$L(\Theta) = f\left(\left\langle \mathbf{Z}^{(\psi_1)} \right\rangle, \ldots, \left\langle \mathbf{Z}^{(\psi_L)} \right\rangle\right)$$

- Parameter shift rule: Computing gradient for SGD by shifting Θ by $\pi/2$

$$\frac{\partial}{\partial \Theta} L(\Theta) = \frac{1}{2}\left(L(\Theta + \frac{\pi}{2}) - L(\Theta - \frac{\pi}{2})\right)$$



- NFT Algorithm: Gradually re-estimating each parameter at timestep t

$$\theta_j^{(t+1)} = -\frac{\pi}{2} - \arctan\frac{y_j^{(t)} - z_j^{(t)}}{2x_j^{(t)} - y_j^{(t)} - z_j^{(t)}}$$

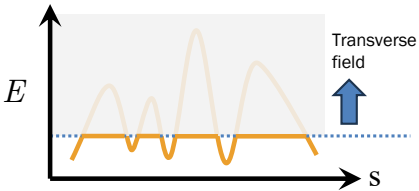
where

$$\begin{aligned} x_j^{(t)} &= L(\Theta^{(t)} | \theta_j = \theta_j^{(t)}) \\ y_j^{(t)} &= L(\Theta^{(t)} | \theta_j = \theta_j^{(t)} + \frac{\pi}{2}) \\ z_j^{(t)} &= L(\Theta^{(t)} | \theta_j = \theta_j^{(t)} - \frac{\pi}{2}) \end{aligned}$$

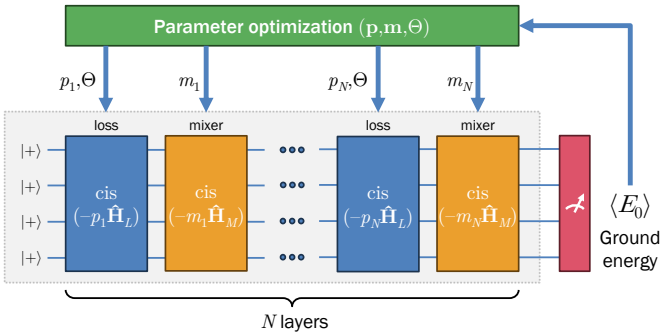
- Both methods converge very fast

Quantum Approximate Optimization Algorithm (QAOA)

- Quantum annealing consists of two steps
  - Mixer: explore the transverse field to improve coverage
  - Loss: identify the globally optimal point in the explored transverse field



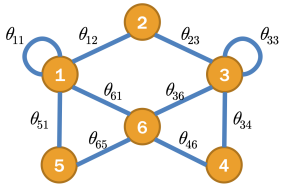
- Objective function = N layers of mixer Hamiltonian $\hat{\mathbf{H}}_M$ + loss Hamiltonian $\hat{\mathbf{H}}_L$



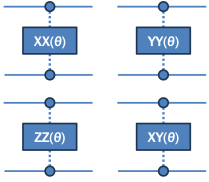
QAOA: Quadratic Unbounded Binary Optimization (QUBO)

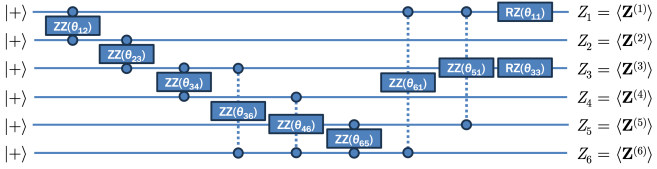
- Each edge (i,j) becomes a model parameter θ_{ij}
- Loss function:

$$L = \sum_{(i,i)\in E} \frac{\theta_{ii}}{2} Z_i + \sum_{(i,j)\in E} \frac{\theta_{ij}}{2} Z_i Z_j$$



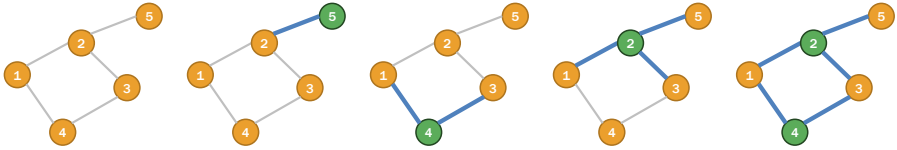
Ising gates rotate both input qubits with the same phase θ





QAOA: Max-Cut as a QUBO Problem

- Graph-coloring problem where the number of edges between two colors is maximum



- Mixer Hamiltonian: X-mixer applies bit-flipping between 0 and 1 on each qubit

$$\hat{\mathrm{H}}_M = \sum_{j=1}^{N} X_j$$

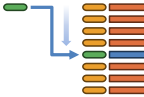
- Loss Hamiltonian: The more adjacent nodes of different colors, the less loss

$$\hat{\mathrm{H}}_L = \sum_{i,j \in E} Z_i Z_j$$

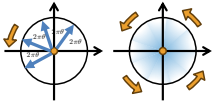
Conclusion

What Problems are Suitable for Quantum Computers?

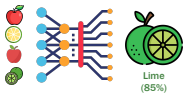
Extreme parallelization



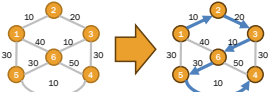
Information search



Phase estimation

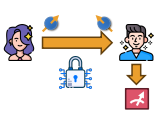


Deep learning with adaptive activation functions

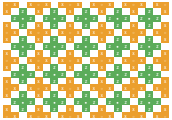


Graph-based optimization

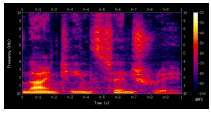
Parallelizable simulation



Secure communication and cryptography



Error correction



Spectral analysis

- Superposition allows quantum computers to perform independent tasks at the same time
- Entanglement fortifies the security of data communication
- The output will be a probability distribution of possible outputs
- Warning: Quantum computers work on complex matrices and eigen-decomposition

Figure 21: Quantum-suitable problems

Questions?